Scientometrics for Success and Influence in the Microsoft Academic Graph

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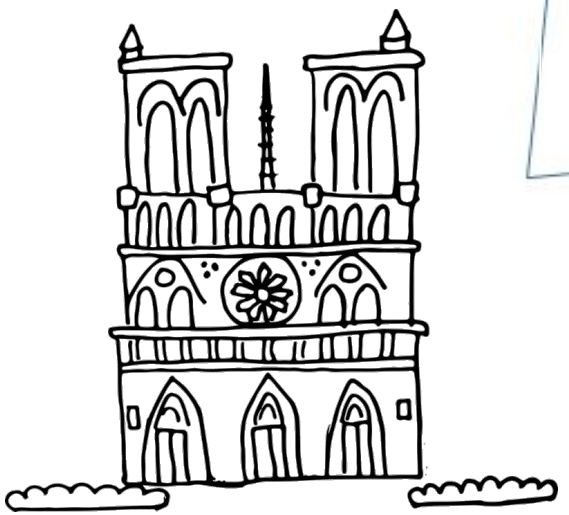
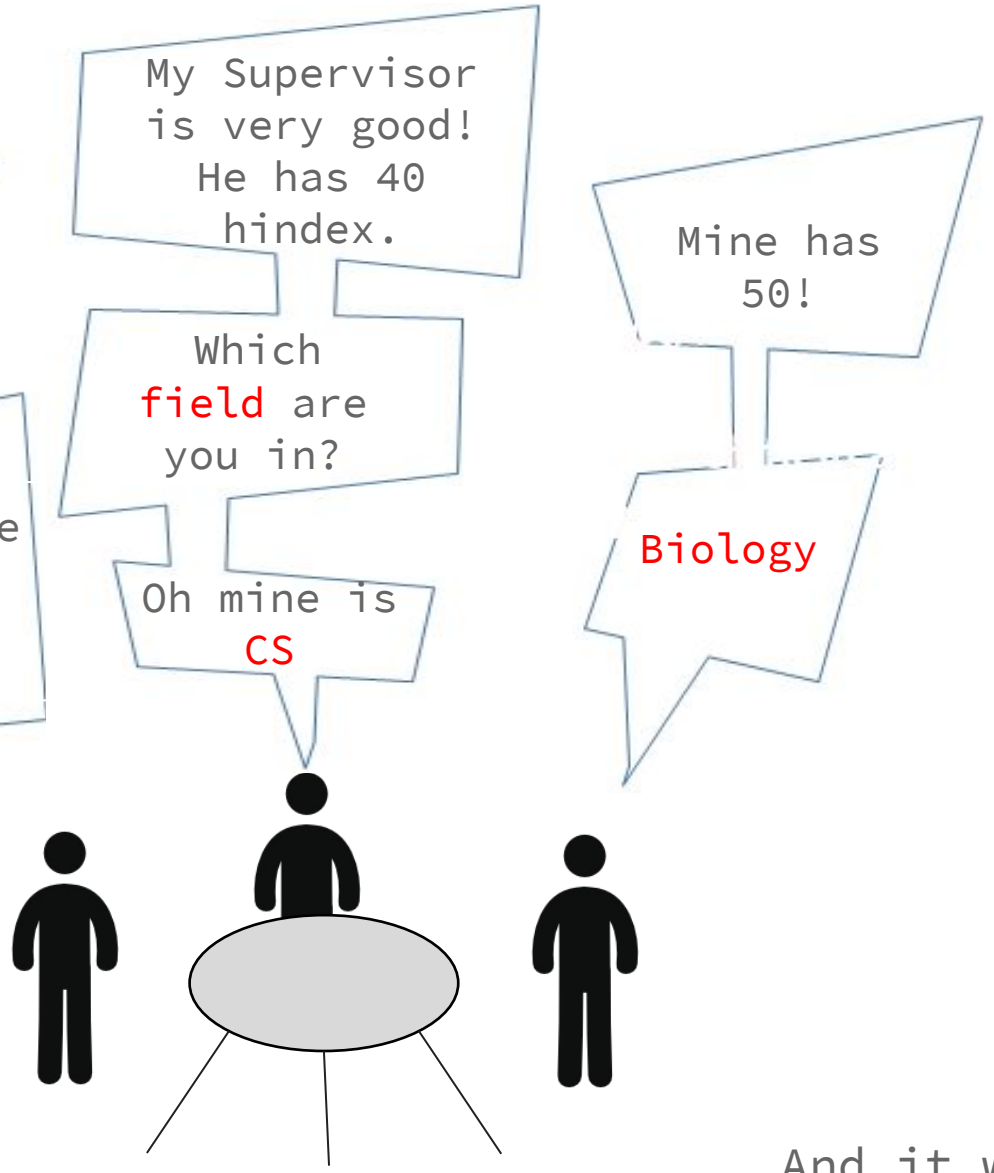
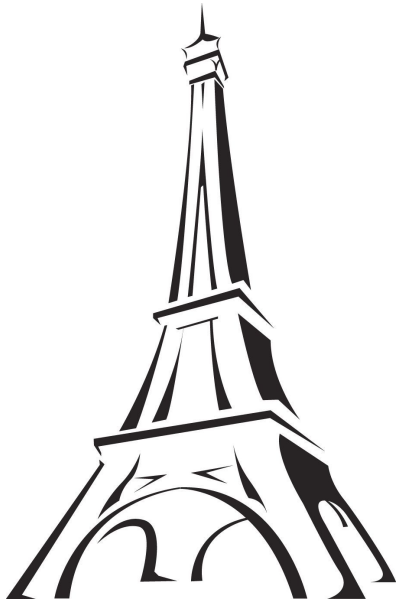
A summer night in Paris...







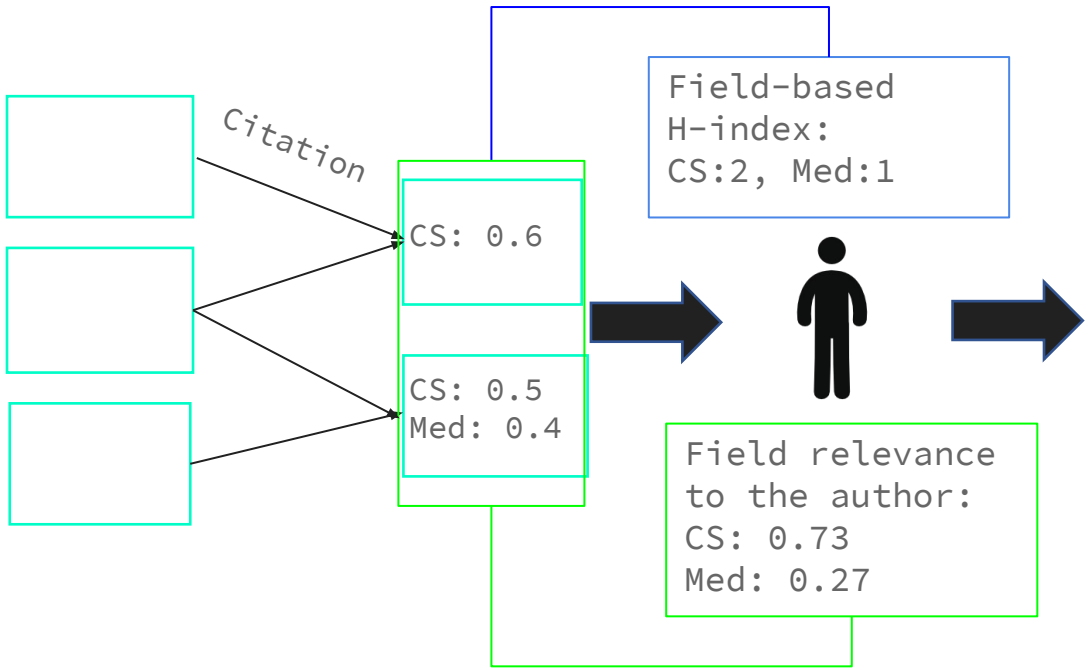
Link to App: http://graphdegeneracy.org/scientometrics/

And it went on like this..

Impact

- Success = Impact and Influence
- Impact is *relative* to the *field*

➡ Use Microsoft Academic Graph (over 200m papers)



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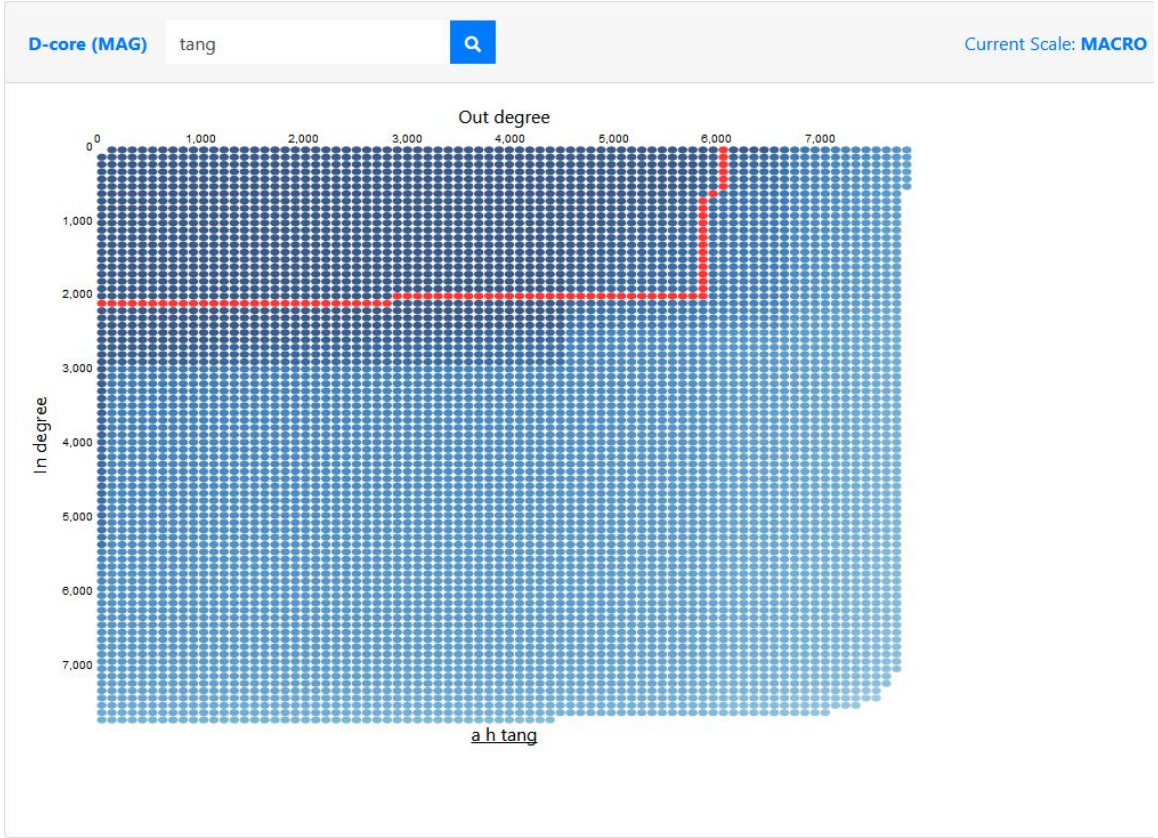
Influence

- K-core in coauthorship?
  - Wrong, I can have many coauthors without being influential
- Use author-citation network
- Problem 1: Direction is important
  - Have to use D-core
  - More complex and less interpretable
- Problem 2: 17.000.000.000 edges
  - Construct it using the egonetworks
  - Remove edges with < 10 citations
  - Run D-core in parallel



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Proof of Concept

- Looked at the authors in the densest D-core
  - These are mostly high energy physicists
  - Their laboratories collaborate a lot with each other -> the papers have a massive number of authors
  - Every new finding cites previous papers with similar authors

The wordcloud of the paper keywords from the authors in the densest d-core.

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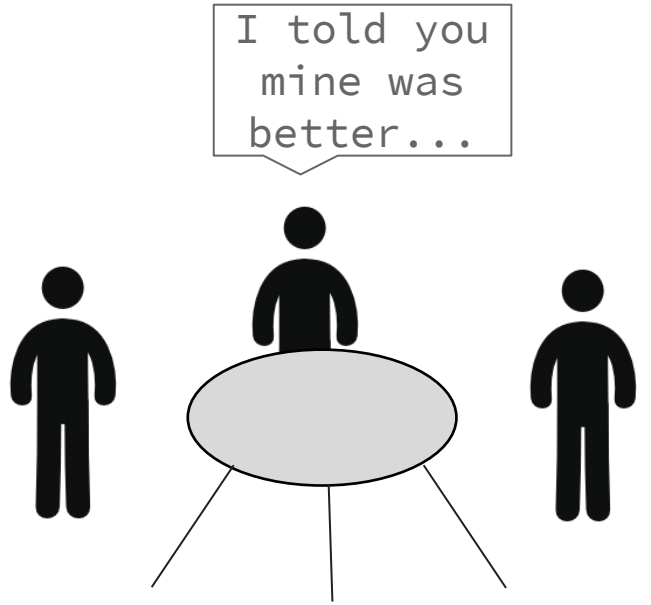
Future work

1. Name disambiguation
2. Self citation rates
3. Number of coauthors



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Thank you

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Code: https://github.com/GiorgosPanagopoulos/MAG-Author-Influence-Scientometrics